

# Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

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**Abstract**—Chemical commands delivered to a living receiver can change how it responds to later commands. Using public experimental FGF2–ERK single-cell trajectories, we ask which information predicts a cellular receiver’s reporter and whether it can shorten repeated use. In nested post hoc comparisons on protocols not used for training, causal reporter filtering reduced the 10-min-ahead error of a current-state predictor by 9.1%, a matched-feedback execution of the authors’ B3 model by a further 4.5%, and input history by a further 1.0%, ending 8.2% below persistence; with leakage-free leave-one-protocol-out selection, input history added 0.1–2.0% in all six folds. For a fixed probe task we define receiver readiness and, in simulation on B3, calibrate waiting rules on half of the posterior samples and evaluate them on the other half. A history-aware rule reached 92.6% success 7.4 min faster than the calibrated fixed wait, but a smoothed reporter did as well, and without observation noise the current reporter sufficed. The waiting gain is filtering, not hidden memory; no result concerns measured task reliability.

**Index Terms**—Molecular communication, cellular receivers, mechanism-informed prediction, receiver reuse, waiting rules, FGF2, ERK.

## I. INTRODUCTION

Molecular communication (MC) with living receivers asks a chemical command to produce a specified response in a cell. Reception then means executing the requested response on time, and a previous command can change the next response through delivery persistence, receptor dynamics, intracellular feedback, or cellular state. For repeated tasks the operational question is when the receiver can be used again at matched reliability. We ground this question in recorded biology: FGF2 stimulation of PC-12 cells with ERK activity read by a FRET reporter [1], treated as a continuous receiver output without retroactive symbols or cell-fate labels.

Two lines of work bound the claim. In cell dynamics, the FGF2 data were used to infer signaling architecture, including feedback and a nontrivial washout response [1]; stimulation frequency shapes ERK responses [2]; automated chemical feedback has controlled mammalian signaling [3]; and a learned dynamical model enabled model predictive control of thousands of single cells [4]. In MC, semantic systems learn task-relevant representations [5], [6], interpretable reaction–diffusion learning has been proposed [7], experimental yeast pheromone links characterize biological receiver noise [8], genetic-circuit receivers are designed computationally [9], calcium signaling with refractory behavior has been controlled

[10], error control has been adapted to cellular noise [11], and long-term experimental MC exposes reuse tradeoffs [12]. Prediction, feedback and predictive control are therefore not new; the open question is which information makes a *next task* predictable and how much a waiting decision gains from it.

This paper makes two bounded contributions. (i) On experimental FGF2–ERK trajectories, a fair nested ablation separates reporter filtering, a calibrated mechanistic increment from the authors’ B3 model, and input history, with leakage-free leave-one-protocol-out (LOPO) validation and phase-wise errors. (ii) For a fixed probe task we define readiness and compare calibrated waiting rules in simulation on B3, with a smoothed-reporter baseline and a noise ablation. Prediction and waiting gains are evaluated separately, and no result concerns measured task reliability.

## II. SYSTEM AND TASK MODEL

Let  $u(t) \geq 0$  denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$c_R(t) = \mathcal{T}_\theta[u](t), \quad \dot{x}_j = f_{\theta_j}(x_j, c_R), \quad y_j = g(x_j) + \nu_j, \quad (1)$$

where  $c_R$  is the local concentration,  $x_j$  the unobserved state of cell  $j$ , and  $y_j$  its normalized reporter. The operator  $\mathcal{T}_\theta$  can include delivery lag and washout; neither it nor  $c_R$  is calibrated here, and the FRET ratio is not a molecular count or a cell-fate measurement. The available information  $\mathcal{H}_t$  contains the executed commands and the observations whose feedback has arrived by  $t$ ; offline, each observation is assumed available at its recorded time.

We fix the amplitude and duration of the next command and decide only when to send it (Fig. 1). After a previous command ending at time 0, a 5-min probe at the same amplitude is sent at  $\tau \in \mathcal{T} = \{0, 2, \dots, 120\}$  min. With  $r_{\text{naive}}$  the receiver’s rise to the same probe without a previous command, the task outcome and readiness are

$$S = \mathbf{1} \left\{ \max_{t \in [\tau, \tau+W]} y(t) - y(\tau) \geq \kappa r_{\text{naive}} \right\}, \quad (2)$$

$$R_t = \mathbb{P}(S = 1 \mid \mathcal{H}_t, \tau = t),$$

with task window  $W = 20$  min and  $\kappa = 0.5$ . The completion time is  $T = \tau + W$  on success and the deadline penalty  $D = 140$  min on failure, and a rule should minimize  $\mathbb{E}[T]$  subject to  $\mathbb{P}(S = 1) \geq q = 0.9$ . Readiness is task-specific: a reporter

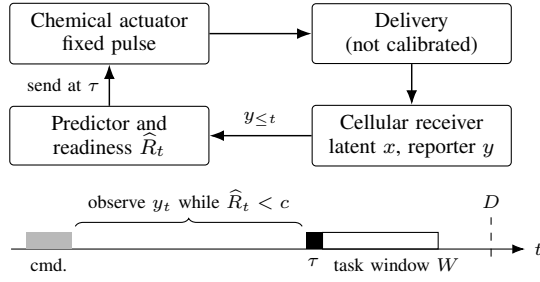


Fig. 1. System and one reuse task. The completed analysis predicts the reporter from commands and observations; delivery and latent states are not calibrated. The waiting rule is evaluated only in simulation on B3 (Sec. V-D).

need not return to baseline before the receiver is ready for this task. The experimental data have no task labels, so Eq. (2) is evaluated only in simulation.

### III. MECHANISM-INFORMED PREDICTION AND WAITING

#### A. Nested Predictors

Separate predictors are fit for horizons  $h = 2$  and 10 min after a fixed 10-min observed warmup. Every learned predictor receives the same known future command (its exposure in 2-min bins, the concentration at  $t + h$ , and  $h$ ); future reporters are targets only. The current-state set  $C$  contains  $u_t$ ,  $u_t^2$ ,  $y_t$ ,  $y_t^2$ ,  $u_t y_t$ , the slope  $(y_t - y_{t-2})/2$  and the future-command features. Causal filters with scales  $\tau \in \{2, 10, 60\}$  min summarize input and reporter history,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (3)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with  $r_{\tau,0} = y_0$  and  $\Delta_i = t_i - t_{i-1}$ ; the scales are computational choices, not identified rate constants.  $O$  adds the reporter filters  $r_\tau$  to  $C$ . As a mechanism we execute the authors' B3 model with their 1,000 posterior parameter samples. With  $f_s(t)$  the noise-free reporter of sample  $s$  under the recorded command, the samples are weighted by each cell's past reporter,

$$w_s(t) \propto \exp \left[ - \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (4)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors.  $M$  adds  $\hat{\delta}_{t,h}$  to  $O$ , and  $H$  adds the input filters  $q_\tau$  and the cumulative exposure  $\int_0^t u(s) ds$  to  $M$ . Persistence  $P$  ( $\hat{y}_{t+h} = y_t$ ) needs no training. Each set is fit by ridge regression with equal weight per condition and standardization from training rows; the penalty (from  $10^{-8}$  to  $10^4$ ) and  $\sigma$  (seven values) are selected on validation. Adding the clock  $t, t^2$  to each set is a sensitivity analysis. The B3 samples express population-mean uncertainty, not cell heterogeneity.

#### B. Calibrated Waiting Rules

A waiting rule probes at

$$\tau = \begin{cases} \inf\{t \in \mathcal{T} : \hat{R}_t \geq c\}, & \text{if such a } t \text{ exists,} \\ \max \mathcal{T}, & \text{otherwise.} \end{cases} \quad (5)$$

TABLE I  
PROTOCOLS, ROLES AND SINGLE-CELL TRAJECTORIES (22 CONDITIONS, 1,565 CELLS, 2-MIN FRAMES). B3 USE: CONCENTRATIONS WHOSE MEANS THE AUTHORS FITTED (FIT) OR USED IN THEIR ARCHITECTURE COMPARISON (ARCH.).

| Protocol      | Role   | ng/ml              | Cells | B3 use         |
|---------------|--------|--------------------|-------|----------------|
| Sustained     | Train  | 0.25, 2.5, 25, 250 | 248   | fit 2.5, 250   |
| 3/20 pulses   | Train  | 2.5, 25, 250       | 207   | fit 2.5, 250   |
| Single 5 min  | Val.   | 0.25, 2.5, 25, 250 | 370   | arch. 2.5, 250 |
| Single 10 min | Test A | 0.25, 2.5, 25, 250 | 248   | —              |
| Single 60 min | Test B | 2.5, 25, 250       | 167   | —              |
| Mixed         | Test C | 0.25, 25           | 171   | —              |
| Mixed         | Viewed | 2.5, 250           | 154   | arch. 2.5, 250 |

Three estimators of  $R_t$  are compared. The current estimator uses only  $y_t$ , through a Gaussian kernel over the (reporter, success) states of reference receivers; the smoothed estimator applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the history-aware estimator weights reference receivers under the known command history by all observations so far, as in Eq. (4). Per-frame probability calibration does not guarantee the reliability of an adaptive stopping rule, so the whole rule is calibrated:  $c$  (and the smoothing scale) is the smallest value whose success reaches  $q$  on calibration receivers, the comparator fixed wait is the shortest wait that does, and success and completion are reported on separate evaluation receivers.

### IV. DATA AND EVALUATION PROTOCOL

#### A. Data and Roles

All numerical files come from the paper-linked `Mijan/LFNS_MSB` repository at pinned commit `5c917ab` [13], with the associated data record [14]; every file matches its Git blob identifier. Commands follow the authors' model XML with half-open pulses; mixed pulses span  $[1, 4)$ ,  $[24, 54)$  and  $[114, 119)$  min, and for this protocol the 10–30 min per-cell normalization was reconstructed from unnormalized exports. Table I lists the 22 conditions. An earlier pilot tested on the viewed mixed blocks, which now serve only as context. The roles were frozen, with a SHA-256 record, before any newly used response file was downloaded. Test A is a new protocol. Test B's  $[0, 60)$  timing was inferred from the authors' B3 simulations, with a  $\pm 1$  min sensitivity, and later confirmed by their simulation summary. Test C uses new concentrations recorded in the session of the viewed blocks. Under that frozen protocol causal history beat a current-state-plus-clock model on Test A by 8.8% but improved on persistence by only 0.7% (one of four concentrations). The nested comparison and the waiting study were each frozen in their own protocol before running, but both are post hoc with respect to inspected data.

#### B. Mechanism, Validation and Scores

B3 is executed by parsing the authors' model files; no equation is retyped, no parameter is refit, and our posterior-predictive mean matches their exported predictions within

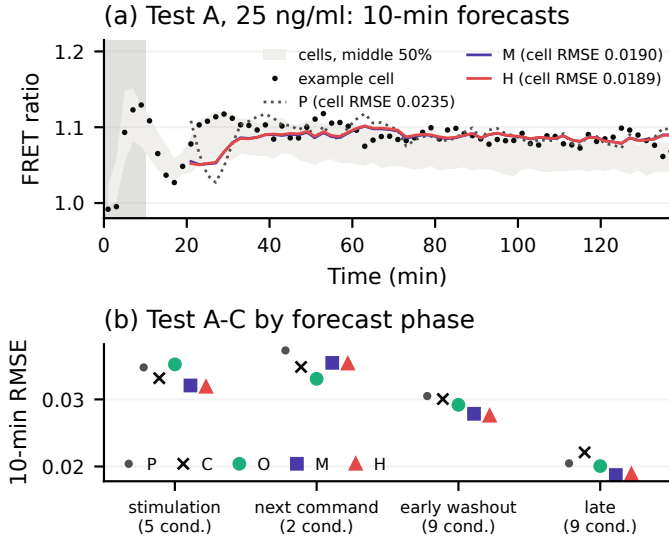


Fig. 2. Experimental 10-min prediction. (a) Test A at 25 ng/ml: the cell with median H error (of 64), its observations, the middle 50% of cells, and the forecasts of P, M and H at their target times; gray: command. (b) Condition-equal RMSE on Tests A–C by forecast phase (axis not starting at zero).

$8.5 \times 10^{-6}$  (acceptance gate  $10^{-3}$ ). Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and viewed mixed blocks at 2.5 and 250 ng/ml (Table I), M and H use a calibrated reference, not a model blind to those protocols. In LOPO each of six protocol groups is held out in turn;  $\sigma$  is selected on the training groups, the penalty by inner leave-one-group-out over the training groups, and the held-out group enters no standardization, selection or fit. Forecast rows are labeled by phase: stimulation (command on at  $t$ ), next command (a pulse starts in  $(t, t + h]$ ), early washout (last pulse ended at most 30 min before  $t$ ), and late. After its warmup Test A contains only washout rows, so it mainly tests washout prediction. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[ \frac{1}{|D|} \sum_{d \in D} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (6)$$

reported with per-condition directions. Rows and cells are not independent interventions and experimental-day identifiers are unavailable, so no significance test is reported.

## V. RESULTS AND DISCUSSION

### A. Nested Prediction

Table II reports the nested predictors. On Test A at 10 min, reporter filtering reduced the error of the current-state set by 9.1% (O versus C), the B3 increment by a further 4.5% (M versus O), and input history by a further 1.0% (H versus M), giving 0.02098, 8.2% below persistence. The mechanistic increment helped in every test role (M versus O: 9.8% on B and 6.0% on C), whereas the filtering gain vanished on B and C. The input-history gain is small and not consistent across conditions: H beat M at two of four concentrations on Test A,

TABLE II  
CONDITION-EQUAL RMSE OF NESTED PREDICTORS AT 10 MIN UNLESS NOTED. VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE; B: SINGLE 60-MIN PULSE (INFERRED TIMING); C: MIXED PULSES AT 0.25 AND 25 NG/ML. P AND C ARE PRE-REGISTERED PREDICTORS; O, M AND H ARE POST HOC. BOLD: SMALLEST PER COLUMN; BOTTOM ROWS COUNT CONDITIONS WITH LOWER MSE.

| Model                 | Val.           | A              | B              | C              | A, 2 min       |
|-----------------------|----------------|----------------|----------------|----------------|----------------|
| P: persistence        | 0.02353        | 0.02287        | 0.02651        | 0.02876        | 0.01302        |
| C: current + slope    | 0.02503        | 0.02442        | 0.02715        | 0.02825        | 0.01342        |
| O: + reporter filters | 0.02403        | 0.02220        | 0.02715        | 0.02827        | 0.01283        |
| M: + B3 increment     | 0.02197        | 0.02120        | 0.02448        | 0.02658        | 0.01263        |
| H: + input history    | <b>0.02195</b> | <b>0.02098</b> | <b>0.02439</b> | <b>0.02656</b> | <b>0.01260</b> |
| H beats M             | 1/4            | 2/4            | 2/3            | 1/2            | 2/4            |
| M beats O             | 4/4            | 3/4            | 2/3            | 2/2            | 3/4            |
| O beats C             | 4/4            | 4/4            | 2/3            | 1/2            | 4/4            |

by 0.4% and 0.1% on B and C, and by 0.2% at 2 min. Clock terms did not help (0.1–2.4% worse on the tests), and C alone was worse than persistence on Tests A and B.

### B. Forecast Phases

In Fig. 2(a) persistence lags the washout by the horizon, whereas M and H follow it (cell RMSE 0.0235, 0.0190 and 0.0189). By phase (Fig. 2(b)), the mechanistic increment reduced the error of O in every phase except the next-command window (by 8.9% during stimulation, 4.6% in early washout and 6.6% late); in early washout O, M and H reached 0.02919, 0.02785 and 0.02757. During stimulation the reporter filters hurt (O 0.03522 versus C 0.03316), and in the late phase M was best with H slightly worse. In the next-command window, covered by two mixed conditions only, O was best and the B3 increment raised the error (0.03544 versus 0.03306), so the calibrated mechanism is least reliable when the next pulse is imminent.

### C. Corrected Leave-One-Protocol-Out

With leakage-free selection, H beat M in all six folds, by 0.1–2.0%, and beat persistence by 8.0–30.6%. The B3 increment contributed more than reporter filtering in five folds (up to 21.6% on 3/20) and was 1.1% worse than O when the 10-min pulse was held out. The earlier frozen LOPO, whose penalties had been selected with the held-out protocols in the training or validation data, put history ahead of persistence by 5.8–13.6%; the corrected comparison keeps that direction. Only the 10- and 60-min folds are untouched by the B3 fit and architecture comparison, whereas the sustained and 3/20 folds contain fitted conditions.

### D. Calibrated Waiting on B3

Readiness and waiting were simulated on B3 with 25 ng/ml commands, a 3- or 30-min previous command, and each posterior sample in turn as the receiver, observed every 2 min with noise of s.d. 0.005. Readiness depended almost only on the current reporter for both histories (Fig. 3(a)). A first task version requiring 80% of the naive response was infeasible for every sample within 120 min and is reported in the

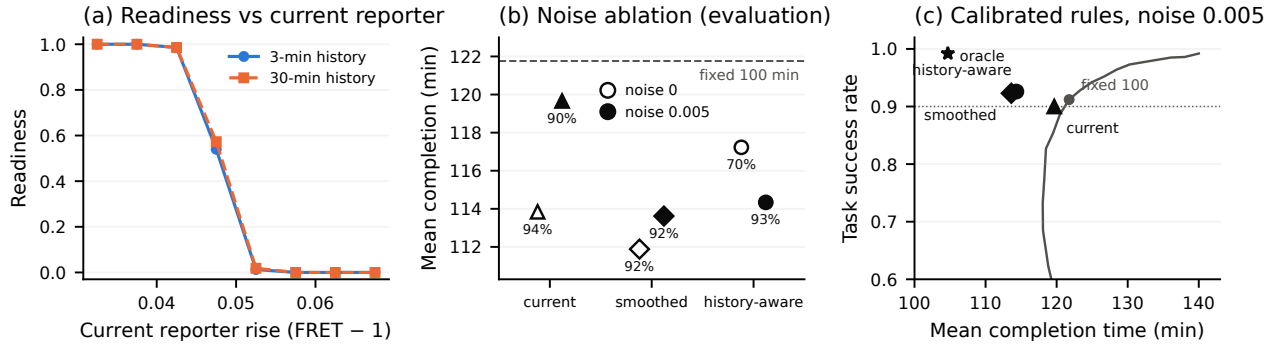


Fig. 3. Calibrated waiting on the authors’ B3 model (25 ng/ml; task: probe rise  $\geq 50\%$  of the naive rise within 20 min). (a) Readiness versus the current noise-free reporter after 3- or 30-min commands. (b) Mean completion on evaluation samples at observation noise 0 (open) and 0.005 (filled), labeled with success; dashed: calibrated fixed wait. (c) Success versus mean completion at noise 0.005; gray: fixed waits; dotted: target 0.9. Thresholds and the fixed wait were chosen on calibration samples. Simulation of one model structure, not measured cells.

package, and the registered candidate waits of 6–48 min never succeeded. With rules calibrated on 500 samples and evaluated on the other 500 (Fig. 3(c)), the history-aware rule reached 92.6% success with a mean completion of 114.3 min, 7.4 min faster than the calibrated fixed wait (100 min: 91.2%, 121.8 min); the current-observation rule reached 90.0% at 119.7 min. A smoothed reporter (16-min scale) did as well as history (92.3%, 113.6 min). Without observation noise (Fig. 3(b)) the current rule reached 93.6% at 113.8 min, whereas the history weights collapsed onto essentially one reference sample (median effective sample size 1.1 versus 28.9 with noise) and the rule missed the calibration target (70.1% on evaluation). The waiting gain is therefore filtering of noisy observations rather than hidden memory; the oracle (99.2%, 104.7 min) bounds what better state estimation could add.

### E. Evidence Boundaries

Seven limits qualify these results. First, history features alone improved little on persistence on the main fresh test (0.7%). Second, the nested and hybrid comparisons are post hoc on inspected roles, not independent confirmation. Third, Test C shares a session with inspected data and no experimental-day identifiers exist, so protocol generalization is not cross-day generalization. Fourth, the waiting results come from B3 simulation: success over posterior samples is not the reliability of a measured cell population, and calibration and evaluation samples share the model structure. Fifth, the waiting gain is attributable to filtering; whether measured cells carry history that matters for control beyond the reporter remains open. Sixth, the task threshold was lowered from 80% to 50% after the first version proved infeasible. Seventh, amplitude and duration were fixed, so nothing is claimed about joint exposure–time optimization. Measured benefits require calibrated delivery, prespecified tasks, and new independent-session experiments.

## VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering and a calibrated mechanistic increment from the authors’ B3

model account for most of the gain in 10-min prediction over current-state and persistence predictors; input history adds a small further gain that holds across protocol folds but not across conditions. In simulation on B3, a calibrated history-aware waiting rule met a 90% target faster than fixed waiting, but no faster than a smoothed reporter.

## DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [13], [14]; author files keep their attribution and receive no new license.

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